

Particle-hole formalism, the normal-ordered Hamiltonian, and full configuration interaction

FYS4480/FYS9480 – Week 37, September 10–11, 2026

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Thursday September 10: Wick's generalised theorem and the Condon-Slater rules; the particle-hole formalism with examples; the normal-ordered Hamiltonian $\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$

Friday September 11: Slater determinants as bit patterns; full configuration interaction theory (chapter 5); the pairing model by FCI; truncations and size consistency; exercises

Reading: these slides, chapter 3 of the lecture notes (sections 3.4–3.5 and 3.13–3.16) and chapter 5; Szabo and Ostlund chapters 2 and 4; Shavitt and Bartlett chapters 3–4. Code: `WeeklySlides/week37.ipynb`, `BookPrograms/chapter03/wick.py`, `BookPrograms/chapter05/fci.py`

Plan for week 37

Thursday September 10, 2×45 min

- Where week 36 stopped: Wick's theorem was proved, the rest was not reached. Short reminder (5 min)
- Wick's generalised theorem for products of normal-ordered groups, proof, and what it buys (sections 3.13–3.14)
- Applications: matrix elements between two-particle states – the Condon-Slater rules (section 3.15)
- The particle-hole formalism: the Fermi sea as vacuum, quasiparticle operators, $\hat{H}_0$ and $\hat{H}_I$ in particle-hole form, with worked examples (section 3.4)
- The normal-ordered Hamiltonian: a proper derivation of $\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$ with $i, j, k, \dots$ below and $a, b, c, \dots$ above the Fermi level (section 3.5)

Friday September 11, 2×45 min

- From the formalism to a running code: Slater determinants as bit patterns, $\hat{H}|\Phi\rangle$ by bit operations (section 3.16)
- Full configuration interaction theory: determinants as a basis, the CI expansion, the eigenvalue problem, the structure of the Hamiltonian matrix, the exponential wall (chapter 5, sections 5.1–5.5)
- FCI for the pairing model, truncated CI, size consistency, the other many-body methods as truncations of FCI, practical considerations (sections 5.6–5.11)
- Exercise session: chapter 3, exercise 4 (normal ordering on a Fermi sea) and chapter 5, exercises 1–3

Not covered: section 5.7 (the Hubbard ring by FCI) – part of next week's examples.

Thursday, lecture 1: Wick's generalised theorem

Where week 36 ended

Last week we established the operators and the theorem that evaluates them:

$$\hat{F} = \sum_{pq} \langle p | \hat{f} | q \rangle a_p^\dagger a_q, \quad \hat{V} = \frac{1}{4} \sum_{pqrs} \langle pq | \hat{v} | rs \rangle_{AS} a_p^\dagger a_q^\dagger a_s a_r,$$
$$xyz \cdots w = N[xyz \cdots w] + \sum_{(1)} N[\cdots] + \sum_{(2)} N[\cdots] + \cdots$$

with the proof by the appending lemma and induction. What we did *not* reach, and do today:

- the **generalised theorem** for products of normal-ordered groups, which is the form in which Wick's theorem is actually used, and the Condon-Slater rules as its first application (sections 3.13–3.15);
- the **particle-hole formalism**: the Fermi sea as vacuum (section 3.4), this time with worked examples;
- the **normal-ordered Hamiltonian** $\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$, derived in full (section 3.5).

Tomorrow the bit representation (section 3.16) and full configuration interaction theory (chapter 5), where all of this becomes a matrix to diagonalise.

Reminder: contractions and the two theorems of week 36

The contraction of two operators is a number, $xy - N[xy] = \langle 0|xy|0\rangle$, and only one type survives with respect to the empty vacuum:

$$\overline{a_p a_q^\dagger} = \delta_{pq}, \quad \overline{a_p a_q} = \overline{a_q^\dagger a_p} = \overline{a_p^\dagger a_q^\dagger} = 0.$$

- **Wick's theorem:** product = normal-ordered product + all single contractions + all double contractions + $\dots$; each term carries the sign $(-1)^\nu$, ν the number of crossings of the contraction lines (equivalently: the parity of the permutation that brings the contracted pairs to the front).
- **Vacuum expectation values:** only fully contracted terms survive, $\langle 0|xyz \dots w|0\rangle = \sum_{\text{complete pairings}} (-1)^\nu \prod(\text{contractions})$; $(M-1)!!$ pairings for M operators, zero for odd M .
- A single contraction of two operators with k operators between them carries the sign $(-1)^k$ – used repeatedly today.

The situation we actually meet

$\langle \Phi | \hat{O} | \Phi' \rangle$: bra, operator and ket, each *already* normal-ordered. Contractions *inside* such a group can be skipped.

Wick's generalised theorem (section 3.13)

Products of normal-ordered groups

$$N[A_1 A_2 \cdots] N[B_1 B_2 \cdots] N[C_1 C_2 \cdots] = N[A_1 A_2 \cdots B_1 B_2 \cdots C_1 C_2 \cdots] + \sum_{\text{all}} N[A_1 A_2 \cdots C_1 C_2 \cdots],$$

where the sum runs over all contractions *between operators belonging to different groups*.

- Contractions within a single group are absent – not thrown away, but **zero**, because that group is already normal-ordered and its internal contractions are of the vanishing types.
- The ordinary theorem is the special case in which every group holds a single operator.
- Taking the vacuum expectation value:

$$\langle 0 | N[A_1 A_2 \cdots] N[B_1 B_2 \cdots] N[C_1 C_2 \cdots] | 0 \rangle = \sum_{\substack{\text{fully contracted,} \\ \text{no line inside a group}}} (-1)^\nu \prod \text{contractions}$$

the form used relentlessly in Hartree-Fock, perturbation theory and coupled cluster theory.

What the restriction buys

Put numbers on it before proving anything (all from `wick.py`, recomputed in section 2 of the notebook):

string	groups	all pairings	inter-group pairings
one-body element $\langle 0 a_j a_i a_p^\dagger a_q a_k^\dagger a_\ell^\dagger 0 \rangle$	$2 + 2 + 2$	$5!! = 15$	8
two-body element $\langle 0 a_j a_i a_p^\dagger a_q^\dagger a_s a_r a_i^\dagger a_j^\dagger 0 \rangle$	$2 + 4 + 2$	$7!! = 105$	24
three two-body operators	$4 + 4 + 4$	$11!! = 10\,395$	1 728

- Cutting a string into groups does *not* tame the factorial (book, Fig. 3.2): the surviving fraction tends to a number fixed by the group *size* – a little under 0.6 for groups of two, about 0.18 for groups of four (0.53 and 0.17 in the table).
- **Fat groups pay:** a matrix element is not $2n$ separate operators but a few normal-ordered blocks, and the bigger the blocks the fewer pairings survive – before a single index has been looked at.
- This is why one always writes the interaction in normal-ordered form *before* starting a derivation – the second half of today.

Proof of the generalised theorem, in three steps (section 3.14)

Step 0: a normal-ordered group has no internal contractions. Each group is *written* with its creation operators to the left (any group can be brought to that form at the price of a sign that multiplies both sides). Pick A_k, A_l with $k < l$: if A_k is a creation operator, A_l is either a creation operator ($a^\dagger a^\dagger \rightarrow 0$) or an annihilation operator to its right ($a^\dagger a \rightarrow 0$); if A_k is an annihilation operator, so is everything to its right ($aa \rightarrow 0$). **The arrangement matters:** written as $a_q a_p^\dagger$ the pair contracts to δ_{pq} .

Step 1: two groups, by induction on the length m of the second group. $m = 1$ is the appending lemma of week 36 with $\Omega = B_1$. For $m \rightarrow m + 1$ multiply by B_{m+1} on the right and apply the lemma to every term: it produces (a) B_{m+1} absorbed into the normal product, (b) contractions of B_{m+1} with the uncontracted operators. A partner from A gives a legitimate A – B line; a partner from $B_1 \dots B_m$ is a line internal to the second group, **zero by step 0**. Every set of A – B lines is generated exactly once, with its crossing sign.

Step 2: any number of groups, by induction on the number of groups: multiplying from the right by one more group $N[H_1 \dots H_k]$ is step 1 with A standing for what is left of the first groups. $\square$

Nothing new was needed

The whole proof is the appending lemma applied over and over, plus step 0. Section 2 of the notebook verifies the theorem as an operator identity for random groups.

Pause to think: the generalised theorem at work

Pause to think #1

(about 4 min – discuss with your neighbour)

1. For the string $\langle 0 | a_j a_i a_p^\dagger a_q a_k^\dagger a_\ell^\dagger | 0 \rangle$ with the three groups indicated, list the pairings that join only different groups. There should be eight. How many of them survive the elementary contractions?
2. A colleague writes the one-body operator as $\sum_{pq} \langle p | \hat{f} | q \rangle a_q a_p^\dagger$ (“it is the same up to a sign and a constant”) and applies the generalised theorem with this as the middle group. What goes wrong?

Answer to pause #1

1. Label positions 1..6: $a_j, a_i \mid a_p^\dagger, a_q \mid a_k^\dagger, a_\ell^\dagger$. Inter-group pairings: $\{(1, 3), (2, 5), (4, 6)\}$, $\{(1, 3), (2, 6), (4, 5)\}$, $\{(1, 4), (2, 5), (3, 6)\}$, $\{(1, 4), (2, 6), (3, 5)\}$, $\{(1, 5), (2, 3), (4, 6)\}$, $\{(1, 5), (2, 4), (3, 6)\}$, $\{(1, 6), (2, 3), (4, 5)\}$, $\{(1, 6), (2, 4), (3, 5)\}$ – eight. A survivor needs every line to join an annihilation operator $(1, 2, 4)$ to a creation operator $(3, 5, 6)$ standing to its right, which kills every pairing containing $(1, 4)$, $(2, 4)$, $(3, 5)$ or $(3, 6)$. Exactly *four* survive: $\{(1, 3), (2, 5), (4, 6)\}$, $\{(1, 3), (2, 6), (4, 5)\}$, $\{(1, 5), (2, 3), (4, 6)\}$, $\{(1, 6), (2, 3), (4, 5)\}$ – one for each of the four terms of the result on the next slides.
2. The middle group is then *not* normal-ordered: its internal contraction $a_q a_p^\dagger \rightarrow \delta_{pq}$ is not zero, and the theorem as stated omits a non-vanishing term ($\sum_p \langle p | \hat{f} | p \rangle = \text{tr } \hat{f}$ times the overlap). Either bring the group to normal order first, $a_q a_p^\dagger = \delta_{pq} - a_p^\dagger a_q$, or include the internal contraction by hand. Step 0 is a hypothesis, not a decoration.

Application 1: a one-body operator between two-particle states (section 3.15)

With $\hat{O}^{(1)} = \sum_{pq} \langle p | \hat{O}^{(1)} | q \rangle a_p^\dagger a_q$ and $|ij\rangle = a_i^\dagger a_j^\dagger |0\rangle$, $\langle ij| = \langle 0 | a_j a_i$,

$$\langle ij | \hat{O}^{(1)} | k\ell \rangle = \sum_{pq} \langle p | \hat{O}^{(1)} | q \rangle \langle 0 | \underbrace{a_j a_i}_{\text{bra}} \underbrace{a_p^\dagger a_q}_{\text{operator}} \underbrace{a_k^\dagger a_\ell^\dagger}_{\text{ket}} | 0 \rangle.$$

Only contractions between different groups contribute, and each must join an annihilation operator to a creation operator standing to its right. One such term:

$$\overbrace{a_j a_i}^{\quad} \overbrace{a_p^\dagger a_q}^{\quad} \overbrace{a_k^\dagger a_\ell^\dagger}^{\quad} = + \delta_{ip} \delta_{qk} \delta_{j\ell} \longrightarrow \delta_{j\ell} \langle i | \hat{O}^{(1)} | k \rangle$$

after the sums over p and q are carried out. No line crosses another: sign $+$.

Application 1: the result

Collecting the four surviving terms (pause #1),

One-body operator between two-particle determinants

$$\langle ij|\hat{O}^{(1)}|k\ell\rangle = \delta_{j\ell}\langle i|\hat{o}^{(1)}|k\rangle - \delta_{jk}\langle i|\hat{o}^{(1)}|\ell\rangle - \delta_{i\ell}\langle j|\hat{o}^{(1)}|k\rangle + \delta_{ik}\langle j|\hat{o}^{(1)}|\ell\rangle.$$

- Diagonal, $k = i, \ell = j$: $\langle i|\hat{o}|i\rangle + \langle j|\hat{o}|j\rangle$ – the sum over occupied orbitals of week 34.
- One orbital replaced, $|k\ell\rangle = |i\ell'\rangle$ with $\ell' \neq j$: a single term, $\langle j|\hat{o}|\ell'\rangle$ (up to the sign of bringing the determinants to maximum coincidence).
- Two orbitals replaced: no assignment of the deltas can be satisfied – **zero**.

These are the **Condon-Slater rules** for one-body operators (week 36, exercise 1), obtained without a single explicit permutation. The same argument for N particles: the bra and ket supply N annihilation and N creation operators, and one contraction must pass through the operator.

Application 2: the two-body interaction, diagonal element

With $\hat{H}_I = \frac{1}{2} \sum_{pqrs} \langle pq|v|rs \rangle a_p^\dagger a_q^\dagger a_s a_r$ (plain elements),

$$\langle ij|\hat{H}_I|ij \rangle = \frac{1}{2} \sum_{pqrs} \langle pq|v|rs \rangle \langle 0| \underbrace{a_j a_i}_{\text{bra}} \underbrace{a_p^\dagger a_q^\dagger a_s a_r}_{\text{operator}} \underbrace{a_i^\dagger a_j^\dagger}_{\text{ket}} |0 \rangle.$$

An eight-operator string: 105 pairings, 24 between different groups, **four** survive. One of them:

$$\overbrace{a_j a_i a_p^\dagger a_q^\dagger} \overbrace{a_s a_r a_i^\dagger a_j^\dagger} = \delta_{ip} \delta_{jq} \delta_{ri} \delta_{sj} \longrightarrow \langle ij|v|ij \rangle,$$

and the one obtained by exchanging the roles of r and s (the lines $(a_s, a_i^\dagger)$ and $(a_r, a_j^\dagger)$), which crosses one more line and enters with the opposite sign: $-\langle ij|v|ji \rangle$. The two remaining terms are these two with $i \leftrightarrow j$ in the operator indices; they are equal to the first two and cancel the factor $\frac{1}{2}$.

Application 2: the result, and the Condon-Slater rules for N particles

Two-body interaction in a two-particle determinant

$$\langle ij|\hat{H}_I|ij\rangle = \langle ij|v|ij\rangle - \langle ij|v|ji\rangle = \langle ij|v|ij\rangle_{\text{AS}}, \quad \text{and likewise} \quad \langle k\ell|\hat{H}_I|ij\rangle = \langle k\ell|v|ij\rangle_{\text{AS}}.$$

For N particles ($\hat{H} = \hat{H}_0 + \hat{V}$, $h_{pq} = \langle p|\hat{h}_0|q\rangle$, AS elements, maximum coincidence):

determinants differ in	$\langle \Phi \hat{H}_0 \Phi'\rangle$	$\langle \Phi \hat{V} \Phi'\rangle$
no orbital	$\sum_i h_{ii}$	$\frac{1}{2} \sum_{ij} \langle ij v ij\rangle$
one orbital, $i \rightarrow a$	h_{ai}	$\sum_j \langle aj v ij\rangle$
two orbitals, $ij \rightarrow ab$	0	$\langle ab v ij\rangle$
three or more	0	0

Many-body connection

An n -body operator connects determinants differing by at most n orbitals: the Hamiltonian matrix in a determinant basis is **banded in the excitation level** and overwhelmingly sparse – tomorrow's FCI matrix. The table's lines become E_{ref} , $\hat{F}_N$, $\hat{V}_N$ of this afternoon's normal-ordered Hamiltonian.

Pause to think: off-diagonal elements

Pause to think #2

(about 4 min – discuss with your neighbour)

Use the generalised theorem to evaluate, for four distinct orbitals i, j, k, ℓ ,

$$\langle k\ell | \hat{H}_I | ij \rangle = \frac{1}{2} \sum_{pqrs} \langle pq | v | rs \rangle \langle 0 | a_\ell a_k a_p^\dagger a_q^\dagger a_s a_r a_i^\dagger a_j^\dagger | 0 \rangle.$$

1. How many inter-group pairings survive, and what is the result?
2. What happens if the determinants differ in only *one* orbital, $\langle kj | \hat{H}_I | ij \rangle$ – and, for N particles, if they differ in three?

Answer to pause #2

1. The bra operators must contract into $a_p^\dagger a_q^\dagger$ (two ways) and $a_s a_r$ into the ket (two ways): four survivors, exactly as for the diagonal element. Result:
 $\langle k\ell | \hat{H}_I | ij \rangle = \langle k\ell | v | ij \rangle - \langle k\ell | v | ji \rangle = \langle k\ell | v | ij \rangle_{AS}$ – a two-body operator connects determinants differing by two orbitals with a single antisymmetrised matrix element.
2. With one orbital in common the four survivors give $\langle kj | v | ij \rangle_{AS}$; for N particles the common orbital can be any occupied m , and the result is $\sum_m \langle km | v | im \rangle_{AS}$ – an *effective one-body* operator, the interaction part of the Fock matrix we meet this afternoon. With three or more differing orbitals at least one bra operator has no creation operator to contract with except those of the ket, and the two-body operator cannot bridge the gap: **zero**. The Hamiltonian matrix is sparse.

Thursday, lecture 2: Particle-hole formalism, normal ordering

No free lunch – and one important exception (section 3.4)

Second quantization is an elegant formalism for constructing states and operators, and expectation values are easy to compute. But from a practical point of view – *solving* the Schrödinger equation – there is no particular gain: the many-body problem is equally hard in any representation.

There is at least one case where second quantization comes to our rescue. It is easy to introduce **another reference state than the empty vacuum**: with many particles present we take a filled determinant

$$|c\rangle \equiv |\alpha_1 \alpha_2 \dots \alpha_{n_c-1} \alpha_{n_c}\rangle = a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger \dots a_{\alpha_{n_c}}^\dagger |0\rangle \quad (c \text{ for core, } n_c \text{ particles})$$

as the new vacuum and describe everything else as *excitations* of it.

- This reduces the dimensionality of the many-body problem considerably, and allows us to sum specific many-body correlations to infinite order (perturbation theory, coupled cluster).
- The states $|c\rangle$, $a_{\alpha_{n_c+1}}^\dagger |c\rangle$, $a_{\alpha_{n_c}} |c\rangle$, $a_{\alpha_{n_c+1}}^\dagger a_{\alpha_{n_c}} |c\rangle$, $\dots$ are the reference state, one-particle, one-hole and one-particle-one-hole states.
- Throughout: $i, j, k, l, \dots$ label states **below** the Fermi level F (holes), $a, b, c, d, \dots$ states **above** it (particles), $p, q, r, s, \dots$ any state. The constraints $\leq F$, $> F$ are implied by the letters and dropped from the sums.

The reference state and its neighbours

In the original representation the $(n_c + 1)$ - and $(n_c - 1)$ -particle states are products of $a^\dagger$'s on $|0\rangle$. Relative to $|c\rangle$,

$$|\alpha_1 \alpha_2 \dots \alpha_{n_c} \alpha_{n_c+1}\rangle = a_{\alpha_1}^\dagger \dots a_{\alpha_{n_c}}^\dagger a_{\alpha_{n_c+1}}^\dagger |0\rangle = (-1)^{n_c} a_{\alpha_{n_c+1}}^\dagger |c\rangle \equiv (-1)^{n_c} |\alpha_{n_c+1}\rangle_c,$$

$$|\alpha_1 \alpha_2 \dots \alpha_{n_c-1}\rangle = a_{\alpha_1}^\dagger \dots a_{\alpha_{n_c-1}}^\dagger |0\rangle = (-1)^{n_c-1} a_{\alpha_{n_c}}^\dagger |c\rangle \equiv (-1)^{n_c-1} |\alpha_{n_c}^{-1}\rangle_c.$$

- The first is a **one-particle state**: one particle added to the reference. The second is a **one-hole state**: one particle removed. The signs are the cost of moving $a_{\alpha_{n_c+1}}^\dagger$ (or $a_{\alpha_{n_c}}$) past the n_c (or $n_c - 1$) creation operators of the core.
- The crucial difference from the true vacuum:

$$a_i |c\rangle \neq 0 \quad \text{for } i \leq F, \quad \text{whereas} \quad a_\alpha |0\rangle = 0 \quad \text{for all } \alpha.$$

An annihilation operator no longer kills the vacuum – so the operators must be redefined.

Quasiparticle operators

We want new operators $b_\alpha^\dagger, b_\alpha$ that behave with respect to $|c\rangle$ exactly as $a^\dagger, a$ did with respect to $|0\rangle$:

$$b_\alpha|c\rangle = 0, \quad \{b_\alpha^\dagger, b_\beta^\dagger\} = \{b_\alpha, b_\beta\} = 0, \quad \{b_\alpha^\dagger, b_\beta\} = \delta_{\alpha\beta}, \quad \langle c|c\rangle = 1.$$

With F the **Fermi level** – the last occupied orbit of $|c\rangle$ – the definition is

Quasiparticle creation and annihilation operators

$$\begin{aligned} b_a^\dagger &= a_a^\dagger, & b_a &= a_a & (a > F), & & b_i^\dagger &= a_i, & b_i &= a_i^\dagger & (i \leq F), \\ b_a^\dagger|c\rangle &= a_a^\dagger|c\rangle = |a\rangle, & b_i^\dagger|c\rangle &= a_i|c\rangle = |i^{-1}\rangle. \end{aligned}$$

- Above F nothing changes. Below F the roles are swapped: *removing* a particle from the core creates a **hole**, and a hole is a quasiparticle.
- The b 's inherit the fermion algebra because swapping $a \leftrightarrow a^\dagger$ for a subset of the labels preserves all three anticommutation relations (pause #3).

Example 1: four orbitals, two particles

Take $n = 4$ orbitals $\alpha_0, \dots, \alpha_3$ and the reference $|c\rangle = a_0^\dagger a_1^\dagger |0\rangle = |1100\rangle$, so $F = 1$: holes $i \in \{0, 1\}$, particles $a \in \{2, 3\}$.

operator	in terms of $a, a^\dagger$	acting on $ c\rangle$
$b_2^\dagger$	$a_2^\dagger$	$a_2^\dagger a_0^\dagger a_1^\dagger 0\rangle = 1110\rangle$, one particle added
$b_1^\dagger$	a_1	$a_1 a_0^\dagger a_1^\dagger 0\rangle = -a_0^\dagger a_1 a_1^\dagger 0\rangle = - 1000\rangle$, one hole
b_2	a_2	0 – orbital 2 is empty
b_1	$a_1^\dagger$	0 – orbital 1 is occupied (Pauli)
$b_2^\dagger b_1^\dagger$	$a_2^\dagger a_1$	$a_2^\dagger (-a_0^\dagger) 0\rangle = + 1010\rangle$: the $1p1h$ state $ 2\ 1^{-1}\rangle$
$b_3^\dagger b_2^\dagger b_1^\dagger b_0^\dagger$	$a_3^\dagger a_2^\dagger a_1 a_0$	$a_3^\dagger a_2^\dagger 0\rangle = - 0011\rangle$: the $2p2h$ state, both particles lifted

- With two particles in four orbitals the $\binom{4}{2} = 6$ determinants are: $|c\rangle$ (no quasiparticle), four $1p1h$ states $|a\ i^{-1}\rangle$, one $2p2h$ state – $1 + 4 + 1 = 6$. Every determinant is an excitation of the reference, and the number of quasiparticles is even for a particle-conserving Hamiltonian.
- $\hat{N} = \sum_p a_p^\dagger a_p$ gives 2 on all six; $\hat{N}_{qp} = \sum_p b_p^\dagger b_p$ gives 0, 2, 4. Section 3 of the notebook does this for $n = 6$, $n_c = 3$, as matrices.

Pause to think: the quasiparticle algebra

Pause to think #3

(about 4 min – discuss with your neighbour)

1. Show, from the definition and the a -algebra, that $\{b_\alpha^\dagger, b_\beta\} = \delta_{\alpha\beta}$ for all four combinations of α, β above or below F , and that $b_\alpha|c\rangle = 0$ for every α .
2. What are $b_i^\dagger b_i|c\rangle$ and $b_i b_i^\dagger|c\rangle$ for a hole state $i \leq F$? Compare with $a_i^\dagger a_i|c\rangle$.

Answer to pause #3

1. Both above F : $\{a_a^\dagger, a_b\} = \delta_{ab}$. Both below: $\{b_i^\dagger, b_j\} = \{a_i, a_j^\dagger\} = \delta_{ij}$. Mixed: $\{b_a^\dagger, b_i\} = \{a_a^\dagger, a_i^\dagger\} = 0$ and $\{b_i^\dagger, b_a\} = \{a_i, a_a\} = 0$ – and $\delta_{ai} = 0$ anyway. For the vacuum: $b_a|c\rangle = a_a|c\rangle = 0$ since a is empty, and $b_i|c\rangle = a_i^\dagger|c\rangle = 0$ since i is occupied (Pauli).
2. $b_i^\dagger b_i|c\rangle = a_i a_i^\dagger|c\rangle = 0$: no quasiparticle in the state i . $b_i b_i^\dagger|c\rangle = a_i^\dagger a_i|c\rangle = |c\rangle$. So $b_i^\dagger b_i$ counts *holes* while $a_i^\dagger a_i = 1 - b_i^\dagger b_i$ counts *particles* in an orbital below F – the origin of the minus sign in the number operator and in the hole term of $\hat{H}_0$ on the next slides.

Particle-hole states, and two number operators

With the new operators we construct many-body quasiparticle states – one-particle-one-hole, two-particle-two-hole, ... – exactly as we constructed many-particle states before:

$$|\beta_1 \beta_2 \dots \beta_{n_p} \gamma_1^{-1} \gamma_2^{-1} \dots \gamma_{n_h}^{-1}\rangle \equiv \underbrace{b_{\beta_1}^\dagger b_{\beta_2}^\dagger \dots b_{\beta_{n_p}}^\dagger}_{>F} \underbrace{b_{\gamma_1}^\dagger b_{\gamma_2}^\dagger \dots b_{\gamma_{n_h}}^\dagger}_{\leq F} |c\rangle.$$

The number operator, rewritten with $a_i^\dagger a_i = b_i b_i^\dagger = 1 - b_i^\dagger b_i$,

$$\hat{N} = \sum_p a_p^\dagger a_p = \sum_a b_a^\dagger b_a + n_c - \sum_i b_i^\dagger b_i, \quad \hat{N} |\beta_1 \dots \gamma_{n_h}^{-1}\rangle = (n_p + n_c - n_h) |\beta_1 \dots \gamma_{n_h}^{-1}\rangle,$$

with n_c the number of particles in $|c\rangle$: $\hat{N}$ counts **particles**. The operator

$$\hat{N}_{qp} = \sum_p b_p^\dagger b_p, \quad \hat{N}_{qp} |\beta_1 \dots \gamma_{n_h}^{-1}\rangle = (n_p + n_h) |\beta_1 \dots \gamma_{n_h}^{-1}\rangle,$$

counts **quasiparticles** – particles above F plus holes below. For a particle-conserving Hamiltonian, $n_p = n_h$ in every state it connects to $|c\rangle$.

The one-body operator in particle-hole form

Split $\hat{H}_0 = \sum_{pq} \langle p | \hat{h}_0 | q \rangle a_p^\dagger a_q$ by p, q above or below F and substitute the b 's. Three blocks are already normal-ordered in the b 's; the hole-hole block needs one anticommutation, $a_i^\dagger a_j = b_i b_j^\dagger = \delta_{ij} - b_j^\dagger b_i$:

$\hat{H}_0$ in particle-hole form

$$\hat{H}_0 = \sum_{ab} \langle a | \hat{h}_0 | b \rangle b_a^\dagger b_b + \sum_{ai} \left[\langle a | \hat{h}_0 | i \rangle b_a^\dagger b_i^\dagger + \langle i | \hat{h}_0 | a \rangle b_i b_a \right] + \sum_i \langle i | \hat{h}_0 | i \rangle - \sum_{ij} \langle j | \hat{h}_0 | i \rangle b_i^\dagger b_j.$$

- First term: acts on particle states only. Last term: on hole states only, with a **minus sign** – a hole in a deep orbital costs energy $-\langle i | \hat{h}_0 | i \rangle$.
- Second term: **creates or destroys a particle-hole pair** – the only piece that connects $|c\rangle$ to other states.
- Third term: a number, the core energy: $\langle c | \hat{H}_0 | c \rangle = \sum_i \langle i | \hat{h}_0 | i \rangle$, every other term having a b on the right or a $b^\dagger$ on the left.

Example 2 (four orbitals, $\hat{h}_0$ diagonal, energies ε_p): $\hat{H}_0 = \sum_a \varepsilon_a b_a^\dagger b_a - \sum_i \varepsilon_i b_i^\dagger b_i + (\varepsilon_0 + \varepsilon_1)$, so $\hat{H}_0 |2\ 1^{-1}\rangle = (\varepsilon_0 + \varepsilon_1 + \varepsilon_2 - \varepsilon_1) |2\ 1^{-1}\rangle = (\varepsilon_0 + \varepsilon_2) |2\ 1^{-1}\rangle$: the orbital energies of the particles present.

✓

The two-body operator in particle-hole form: five groups

Translating the four indices of $\hat{H}_I = \frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs\rangle_{AS} a_p^\dagger a_q^\dagger a_s a_r$ into particle and hole labels and normal-ordering in the b 's, one regroups $\hat{H}_I = \hat{H}_I^{(a)} + \dots + \hat{H}_I^{(e)}$ by the number of particle indices (antisymmetrised elements; section 5 of the notebook checks the decomposition as a matrix identity):

$$\hat{H}_I^{(a)} = \frac{1}{4} \sum_{abcd} \langle ab|\hat{v}|cd\rangle b_a^\dagger b_b^\dagger b_d b_c \quad \text{particle-particle scattering}$$

$$\hat{H}_I^{(b)} = \frac{1}{2} \sum_{abci} \left[\langle ab|\hat{v}|ci\rangle b_a^\dagger b_b^\dagger b_i^\dagger b_c + \langle ci|\hat{v}|ab\rangle b_c^\dagger b_i b_b b_a \right] \quad 1p \rightarrow 2p1h \text{ and back}$$

$$\begin{aligned} \hat{H}_I^{(c)} = \frac{1}{4} \sum_{abij} \left[\langle ab|\hat{v}|ij\rangle b_a^\dagger b_b^\dagger b_j^\dagger b_i^\dagger + \langle ij|\hat{v}|ab\rangle b_i b_j b_b b_a \right] & \quad 2p2h \text{ creation/destruction} \\ + \sum_{abij} \langle ai|\hat{v}|bj\rangle b_a^\dagger b_j^\dagger b_b b_i + \sum_{abi} \langle ai|\hat{v}|bi\rangle b_a^\dagger b_b & \quad \text{ph scattering; particle energy} \end{aligned}$$

$$\hat{H}_I^{(d)} = \frac{1}{2} \sum_{aijk} \left[\langle ai|\hat{v}|jk\rangle b_a^\dagger b_k^\dagger b_j^\dagger b_i + \langle jk|\hat{v}|ai\rangle b_i^\dagger b_j b_k b_a \right] + \sum_{aij} \langle ai|\hat{v}|ji\rangle \left[b_a^\dagger b_j^\dagger + b_j b_a \right] \quad 1h \rightarrow 1p2h; \text{ ph pair}$$

$$\hat{H}_I^{(e)} = \frac{1}{4} \sum_{ijkl} \langle kl|\hat{v}|ij\rangle b_i^\dagger b_j^\dagger b_l b_k - \sum_{ijk} \langle ij|\hat{v}|kj\rangle b_k^\dagger b_i + \frac{1}{2} \sum_{ij} \langle ij|\hat{v}|ij\rangle \quad \text{hh; hole energy; core energy}$$

Each term conserves the particle number and changes the quasiparticle number by $-4, -2, 0, +2, +4$; prefactors $\frac{1}{4}, \frac{1}{2}, 1$ for zero, one, two mixed index pairs. By brute force – sixteen index combinations, anticommutators in each – this is painful and error-prone. The rest of the lecture obtains the same content in four lines with Wick's theorem, without even needing the b 's.

Wick's theorem relative to the Fermi vacuum (section 3.5)

Nothing in the proofs of week 36 used that $|0\rangle$ is empty. Replace $|0\rangle \rightarrow |c\rangle$, $a, a^\dagger \rightarrow b, b^\dagger$: normal ordering, contractions, both theorems and their proofs carry over word for word. We write $\{\dots\}$ for **normal ordering with respect to $|c\rangle$** : quasiparticle creators $a_a^\dagger, a_i$ to the left, annihilators $a_a, a_i^\dagger$ to the right, with the sign of the permutation. Then $\langle c|\{\text{anything}\}|c\rangle = 0$, and the contractions $xy - \{xy\} = \langle c|xy|c\rangle$ are

$$\overline{a_a a_b^\dagger} = \delta_{ab} \quad (a, b > F), \quad \overline{a_i^\dagger a_j} = \delta_{ij} \quad (i, j \leq F), \quad \text{all others zero.}$$

In one formula, with $n_p = 1$ for $p \leq F$ and $n_p = 0$ for $p > F$

$$\overline{a_p^\dagger a_q} = \delta_{pq} n_p \equiv \delta_{pq}^h, \quad \overline{a_p a_q^\dagger} = \delta_{pq}(1 - n_p), \quad \overline{a_p a_q} = \overline{a_p^\dagger a_q^\dagger} = 0.$$

- The second contraction is the new one: a hole state is occupied in $|c\rangle$, so removing and replacing a particle there gives unity.
- $\{a_a^\dagger a_b\} = b_a^\dagger b_b$, $\{a_i^\dagger a_j\} = -a_j a_i^\dagger = -b_j^\dagger b_i$, $\{a_a^\dagger a_i\} = b_a^\dagger b_i^\dagger$, $\{a_i^\dagger a_a\} = b_i b_a$: the bracket is the b -language without renaming operators (section 4 of the notebook).

The normal-ordered one-body operator

Wick's theorem for two operators, relative to $|c\rangle$:

$$a_p^\dagger a_q = \overline{a_p^\dagger a_q} + a_p^\dagger a_q = \{a_p^\dagger a_q\} + \delta_{pq} n_p.$$

Multiply by $\langle p|\hat{h}_0|q\rangle$ and sum; the delta with n_p restricts the constant to hole states:

One-body operator relative to the reference

$$\hat{H}_0 = \sum_{pq} \langle p|\hat{h}_0|q\rangle a_p^\dagger a_q = \underbrace{\sum_i \langle i|\hat{h}_0|i\rangle}_{\langle c|\hat{H}_0|c\rangle} + \sum_{pq} \langle p|\hat{h}_0|q\rangle \{a_p^\dagger a_q\}.$$

- Splitting p, q into a, i and using the dictionary of the previous slide reproduces the four-term particle-hole form – including the minus sign of the hole term – in one line.
- The general pattern: **operator = its expectation value in $|c\rangle$ + normal-ordered remainder**.
The remainder has zero expectation value in $|c\rangle$ and is what acts on excitations.

The two-body string: all contractions relative to $|c\rangle$

For $a_p^\dagger a_q^\dagger a_s a_r$ (positions 1, 2, 3, 4) a non-zero contraction joins a creation operator to an annihilation operator *to its right* with a common hole index – four candidates (1, 3), (1, 4), (2, 3), (2, 4) – and a single contraction carries $(-1)^k$, k the number of operators between the pair:

$$\overbrace{a_p^\dagger a_q^\dagger a_s a_r} = -\delta_{ps}^h \{a_q^\dagger a_r\}, \quad \overbrace{a_p^\dagger a_q^\dagger a_s a_r} = +\delta_{pr}^h \{a_q^\dagger a_s\}, \quad \overbrace{a_p^\dagger a_q^\dagger a_s a_r} = +\delta_{qs}^h \{a_p^\dagger a_r\}, \quad \overbrace{a_p^\dagger a_q^\dagger a_s a_r} = -\delta_{qr}^h \{a_p^\dagger a_s\}.$$

The two double contractions follow the crossing rule – (1, 3)(2, 4) crosses once, (1, 4)(2, 3) is nested:

$$\overbrace{\overbrace{a_p^\dagger a_q^\dagger a_s a_r}} = -\delta_{ps}^h \delta_{qr}^h, \quad \overbrace{\overbrace{a_p^\dagger a_q^\dagger a_s a_r}} = +\delta_{pr}^h \delta_{qs}^h.$$

Wick's theorem for the two-body string, relative to $|c\rangle$ (an operator identity, checked in the notebook for all 6^4 index choices)

$$a_p^\dagger a_q^\dagger a_s a_r = \{a_p^\dagger a_q^\dagger a_s a_r\} - \delta_{ps}^h \{a_q^\dagger a_r\} + \delta_{pr}^h \{a_q^\dagger a_s\} + \delta_{qs}^h \{a_p^\dagger a_r\} - \delta_{qr}^h \{a_p^\dagger a_s\} + \delta_{pr}^h \delta_{qs}^h - \delta_{ps}^h \delta_{qr}^h.$$

Collecting the one-body pieces

Insert the expansion into $\hat{H}_I = \frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs\rangle a_p^\dagger a_q^\dagger a_s a_r$ (antisymmetrised elements). Each singly contracted term carries one hole delta, turning one of the four sums into a sum over holes:

$$\begin{aligned} -\frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs\rangle \delta_{ps}^h \{a_q^\dagger a_r\} &= -\frac{1}{4} \sum_{iqr} \langle iq|\hat{v}|ri\rangle \{a_q^\dagger a_r\} = +\frac{1}{4} \sum_{iqr} \langle qi|\hat{v}|ri\rangle \{a_q^\dagger a_r\}, \\ +\frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs\rangle \delta_{pr}^h \{a_q^\dagger a_s\} &= +\frac{1}{4} \sum_{iqs} \langle iq|\hat{v}|is\rangle \{a_q^\dagger a_s\} = +\frac{1}{4} \sum_{iqs} \langle qi|\hat{v}|si\rangle \{a_q^\dagger a_s\}, \\ +\frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs\rangle \delta_{qs}^h \{a_p^\dagger a_r\} &= +\frac{1}{4} \sum_{pir} \langle pi|\hat{v}|ri\rangle \{a_p^\dagger a_r\}, \\ -\frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs\rangle \delta_{qr}^h \{a_p^\dagger a_s\} &= -\frac{1}{4} \sum_{pis} \langle pi|\hat{v}|is\rangle \{a_p^\dagger a_s\} = +\frac{1}{4} \sum_{pis} \langle pi|\hat{v}|si\rangle \{a_p^\dagger a_s\}, \end{aligned}$$

using $\langle pq|\hat{v}|rs\rangle = -\langle pq|\hat{v}|sr\rangle = -\langle qp|\hat{v}|rs\rangle = \langle qp|\hat{v}|sr\rangle$. After renaming the summation indices all four are identical, and the $\frac{1}{4}$ is cancelled:

one-body part of $\hat{H}_I = \sum_{pq} \left[\sum_i \langle pi|\hat{v}|qi\rangle \right] \{a_p^\dagger a_q\}$

the interaction of one particle (or hole) with all particles of the core – the signs from antisymmetry and from the contraction rule cancelled in pairs.

The constant, and the result

The two doubly contracted terms are numbers,

$$\frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs \rangle [\delta_{pr}^h \delta_{qs}^h - \delta_{ps}^h \delta_{qr}^h] = \frac{1}{4} \sum_{ij} [\langle ij|\hat{v}|ij \rangle - \langle ij|\hat{v}|ji \rangle] = \frac{1}{2} \sum_{ij} \langle ij|\hat{v}|ij \rangle,$$

the direct-minus-exchange energy of the core (antisymmetrised elements: the two terms are equal). Adding $\hat{H}_0$:

The normal-ordered Hamiltonian

$$\begin{aligned} \hat{H} &= E_{\text{ref}} + \hat{F}_N + \hat{V}_N, & E_{\text{ref}} &= \sum_i \langle i|\hat{h}_0|i \rangle + \frac{1}{2} \sum_{ij} \langle ij|\hat{v}|ij \rangle, \\ \hat{F}_N &= \sum_{pq} f_{pq} \{ a_p^\dagger a_q \}, & f_{pq} &= \langle p|\hat{h}_0|q \rangle + \sum_i \langle pi|\hat{v}|qi \rangle, & \hat{V}_N &= \frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs \rangle \{ a_p^\dagger a_q^\dagger a_s a_r \}. \end{aligned}$$

- $E_{\text{ref}} = \langle c|\hat{H}|c \rangle$ is the energy functional of week 34 – there from $N!$ permutations, here from two contractions. f_{pq} is the **Fock matrix**: $\hat{h}_0$ dressed by the mean field of the core; its diagonalisation is Hartree-Fock theory (end of next week).
- $\hat{V}_N$ has the matrix elements of $\hat{V}$; only the string is normal-ordered relative to $|c \rangle$. Section 4 of the notebook builds the three pieces as matrices: $\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$ to machine precision.

Reading the result: the particle and hole blocks

Splitting every index into a or i and translating $\{\dots\}$ into b 's recovers the particle-hole forms of section 3.4 – with the prefactors now *derived*:

$$\hat{F}_N = \sum_{ab} f_{ab} b_a^\dagger b_b + \sum_{ai} [f_{ai} b_a^\dagger b_i^\dagger + f_{ia} b_i b_a] - \sum_{ij} f_{ji} b_i^\dagger b_j,$$

$$\begin{aligned} \hat{V}_N = & \frac{1}{4} \sum_{abcd} \langle ab | \hat{v} | cd \rangle b_a^\dagger b_b^\dagger b_d b_c + \frac{1}{2} \sum_{abci} [\langle ab | \hat{v} | ci \rangle b_a^\dagger b_b^\dagger b_i^\dagger b_c + \text{h.c.}] + \frac{1}{4} \sum_{abij} [\langle ab | \hat{v} | ij \rangle b_a^\dagger b_b^\dagger b_j^\dagger b_i^\dagger + \text{h.c.}] \\ & + \sum_{abij} \langle ai | \hat{v} | bj \rangle b_a^\dagger b_j^\dagger b_b b_i + \frac{1}{2} \sum_{aijk} [\langle ai | \hat{v} | jk \rangle b_a^\dagger b_k^\dagger b_j^\dagger b_i + \text{h.c.}] + \frac{1}{4} \sum_{ijkl} \langle kl | \hat{v} | ij \rangle b_i^\dagger b_j^\dagger b_l b_k. \end{aligned}$$

- The one-body pieces of the five groups, $\sum_{abi} \langle ai | \hat{v} | bi \rangle b_a^\dagger b_b$, $-\sum_{ijk} \langle ij | \hat{v} | kj \rangle b_k^\dagger b_i$ and $\sum_{aij} \langle ai | \hat{v} | ji \rangle (b_a^\dagger b_j^\dagger + b_j b_a)$, are exactly the interaction part of $\hat{F}_N$: they have been **absorbed into the Fock matrix**. That is the whole point of normal ordering.
- Prefactor rule, now explained: restricting $\frac{1}{4} \sum_{pqrs}$ to letters of fixed type keeps the $\frac{1}{4}$ when both pairs are of one kind (ab, ij). A mixed pair such as $(r, s) = (c, i)$ occurs twice in the unrestricted sum, as (c, i) and (i, c) , with equal contributions by antisymmetry: $\frac{1}{4} \cdot 2 = \frac{1}{2}$ for one mixed pair, $\frac{1}{4} \cdot 2 \cdot 2 = 1$ for two.
- Everything on this slide is verified in section 5 of the notebook by comparing matrices.

Pause to think: reading off matrix elements

Pause to think #4

(about 4 min – discuss with your neighbour)

With $\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$ and the generalised theorem relative to $|c\rangle$:

1. Why is $\langle c|\hat{H}|c\rangle = E_{\text{ref}}$ immediate, without any calculation?
2. Consider $\langle \Phi_i^a|\hat{H}|c\rangle$ with $|\Phi_i^a\rangle = a_a^\dagger a_i |c\rangle$, so $\langle \Phi_i^a| = \langle c|a_i^\dagger a_a$. Which of the three pieces can contribute, and how many contraction patterns survive in each?

Answer to pause #4

1. $\hat{F}_N$ and $\hat{V}_N$ are normal-ordered relative to $|c\rangle$, so $\langle c|\hat{F}_N|c\rangle = \langle c|\hat{V}_N|c\rangle = 0$ – a quasiparticle annihilator on the right, or a creator on the left, kills the reference. Only the number E_{ref} is left. (The same argument in the b -language: every term with at least one b or $b^\dagger$ has zero expectation value in $|c\rangle$.)
2. $E_{\text{ref}}\langle c|a_i^\dagger a_a|c\rangle = 0$: $a_a|c\rangle = 0$. In $\hat{V}_N$ the bra supplies two operators and the group $\{a_p^\dagger a_q^\dagger a_s a_r\}$ four; two of the four must be contracted with each other, which the generalised theorem forbids – **zero**. In $\hat{F}_N$ the bra pair $a_i^\dagger a_a$ and the group $\{a_p^\dagger a_q\}$ can be fully contracted in exactly *one* way: a_a with $a_p^\dagger$ (δ_{ap} , particle) and $a_i^\dagger$ with a_q (δ_{iq} , hole), nested lines, sign $+$. Result: $\langle \Phi_i^a | \hat{H} | c \rangle = f_{ai}$. Next slide.

Example 3: the reference coupled to single and double excitations

$$\langle \Phi_i^a | \hat{H} | c \rangle = \sum_{pq} f_{pq} \langle c | \overbrace{a_i^\dagger a_a a_p^\dagger a_q} | c \rangle = \sum_{pq} f_{pq} \delta_{ap} \delta_{iq} = \boxed{f_{ai}}$$

- **Brillouin's theorem:** if the orbitals are chosen such that the Fock matrix is diagonal – the Hartree-Fock orbitals – then $f_{ai} = 0$ and the reference does not couple to any $1p1h$ state. This is what “Hartree-Fock” will mean at the end of next week.

For $|\Phi_{ij}^{ab}\rangle = a_a^\dagger a_b^\dagger a_j a_i |c\rangle$, $\langle \Phi_{ij}^{ab} | = \langle c | a_i^\dagger a_j^\dagger a_b a_a$, only $\hat{V}_N$ can contribute (four bra operators need four partners); a_a, a_b contract with $a_p^\dagger, a_q^\dagger$ in two ways and $a_i^\dagger, a_j^\dagger$ with a_s, a_r in two ways, the four terms being equal by antisymmetry:

$$\langle \Phi_{ij}^{ab} | \hat{H} | c \rangle = \frac{1}{4} \sum_{pqrs} \langle pq | \hat{v} | rs \rangle \langle c | a_i^\dagger a_j^\dagger a_b a_a \{ a_p^\dagger a_q^\dagger a_s a_r \} | c \rangle = \boxed{\langle ab | \hat{v} | ij \rangle}, \quad \langle \Phi_{ijk}^{abc} | \hat{H} | c \rangle = 0.$$

- The three boxed results – E_{ref} , f_{ai} , $\langle ab | \hat{v} | ij \rangle$, and zero beyond – are the first row of tomorrow's FCI matrix, and the Condon-Slater table of lecture 1 read in the new language.
- Signs: the notebook checks all three against the matrices for every choice of a, b, i, j (section 5).

Example 4: the energies of excited determinants

Diagonal elements need all three pieces. For $\langle \Phi_i^a | \hat{H} | \Phi_i^a \rangle = \langle c | a_i^\dagger a_a (E_{\text{ref}} + \hat{F}_N + \hat{V}_N) a_a^\dagger a_i | c \rangle$: E_{ref} times the norm $\langle c | a_i^\dagger a_a a_a^\dagger a_i | c \rangle = 1$; in $\hat{F}_N$ the group $\{a_p^\dagger a_q\}$ contracts either with the two particle operators ($+f_{aa}$) or with the two hole operators (an odd number of crossings: $-f_{ii}$); in $\hat{V}_N$ all four operators must be contracted outwards, one particle line in and out, one hole line in and out: $-\langle ai | \hat{v} | ai \rangle$.

Energies of a $1p1h$ and of a $2p2h$ determinant

$$\begin{aligned} \langle \Phi_i^a | \hat{H} | \Phi_i^a \rangle &= E_{\text{ref}} + f_{aa} - f_{ii} - \langle ai | \hat{v} | ai \rangle, \\ \langle \Phi_{ij}^{ab} | \hat{H} | \Phi_{ij}^{ab} \rangle &= E_{\text{ref}} + f_{aa} + f_{bb} - f_{ii} - f_{jj} + \langle ab | \hat{v} | ab \rangle + \langle ij | \hat{v} | ij \rangle - \sum_{c \in \{a,b\}} \sum_{k \in \{i,j\}} \langle ck | \hat{v} | ck \rangle. \end{aligned}$$

- Read: a particle costs f_{aa} , a hole $-f_{ii}$; the corrections remove the interaction with the *missing* core particle that f_{aa} still counts (and, for $2p2h$, add the interaction of the two new particles and of the two holes).
- Off-diagonal: $\langle \Phi_i^a | \hat{H} | \Phi_j^b \rangle = \delta_{ij} f_{ab} - \delta_{ab} f_{ji} + \langle aj | \hat{v} | ib \rangle$ – the **Tamm-Dancoff** matrix, later in the course. All verified in section 5 of the notebook.

Example 5: the pairing model in particle-hole language

Four doubly degenerate levels $p = 1, \dots, 4$, energies $\varepsilon_p = (p - 1)\xi$, spin $\sigma = \pm$, four particles, and

$$\hat{H} = \xi \sum_{p\sigma} (p - 1) a_{p\sigma}^\dagger a_{p\sigma} - \frac{g}{2} \sum_{pq} a_{p+}^\dagger a_{p-}^\dagger a_{q-} a_{q+}, \quad \text{i.e.} \quad \langle p+ p- | \hat{v} | q+ q- \rangle = -\frac{g}{2}, \text{ else } 0.$$

Reference $|c\rangle$: levels 1 and 2 filled ($n_c = 4$, holes $1\pm, 2\pm$, particles $3\pm, 4\pm$). Then, with $\xi = 1$:

quantity	formula	value
E_{ref}	$\sum_i \varepsilon_i + \frac{1}{2} \sum_{ij} \langle ij \hat{v} ij \rangle = 2\xi + \frac{1}{2} \cdot 4 \cdot (-\frac{g}{2})$	$2 - g$
f_{ii} , hole	$\varepsilon_i + \sum_j \langle ij \hat{v} ij \rangle = \varepsilon_i - \frac{g}{2}$ (only the partner $\bar{i}$ contributes)	$-\frac{g}{2}, 1 - \frac{g}{2}$
f_{aa} , particle	$\varepsilon_a + \sum_j \langle aj \hat{v} aj \rangle = \varepsilon_a$ (no hole partner)	$2, 3$
f_{ai}	0: f is diagonal, Brillouin's theorem holds automatically	0
$\langle \Phi_{1+1-}^{3+3-} \hat{H} c \rangle$	$\langle 3+3- \hat{v} 1+1- \rangle$, a pair moved as a whole	$-\frac{g}{2}$
$\langle \Phi_{1+2+}^{3+3-} \hat{H} c \rangle$	$\langle 3+3- \hat{v} 1+2+ \rangle$, a broken pair	0
$\langle \Phi_{1+1-}^{3+3-} \hat{H} \Phi_{1+1-}^{3+3-} \rangle$	$E_{\text{ref}} + 2\varepsilon_3 - 2(\varepsilon_1 - \frac{g}{2}) + (-\frac{g}{2}) + (-\frac{g}{2}) - 0$	$6 - g$

- The last line is the diagonal element $2\xi[(2 - 1) + (3 - 1)] - g$ of the configuration with pairs in levels 2 and 3: tomorrow's 6×6 matrix, entry by entry (section 6 of the notebook prints every number). The interaction never breaks a pair: $\hat{H}$ connects $|c\rangle$ to four $2p2h$ states only, each with $-g/2$.

Summary of Thursday, and what comes on Friday

Today

- Generalised theorem: only inter-group lines; $105 \rightarrow 24 \rightarrow 4$ for a two-body matrix element
- Condon-Slater: $\sum_i h_{ii} + \frac{1}{2} \sum \langle ij | v | ij \rangle$, $h_{ai} + \sum_j \langle aj | v | ij \rangle$, $\langle ab | v | ij \rangle$, 0
- Fermi sea as vacuum: $b_a^\dagger = a_a^\dagger$, $b_i^\dagger = a_i$; two contractions, $a_a a_b^\dagger \rightarrow \delta_{ab}$ and $a_i^\dagger a_j \rightarrow \delta_{ij}$
- $\hat{H} = E_{\text{ref}} + \sum f_{pq} \{ a_p^\dagger a_q \} + \frac{1}{4} \sum \langle pq | v | rs \rangle \{ a_p^\dagger a_q^\dagger a_s a_r \}$
- $\langle \Phi_i^a | \hat{H} | c \rangle = f_{ai}$, $\langle \Phi_{ij}^{ab} | \hat{H} | c \rangle = \langle ab | v | ij \rangle$

Friday

- Slater determinants as bit patterns; $\hat{H}|\Phi\rangle$ by bit operations
- Full configuration interaction: the determinant basis, the CI expansion, the eigenvalue problem
- The structure of the Hamiltonian matrix and the exponential wall
- FCI for the pairing model: 70 determinants versus 6
- Truncations, size consistency, the map of many-body methods; exercises

Before Friday: run sections 3–6 of `week37.ipynb`, and redo Example 5 for the reference with levels 1 and 3 filled – which matrix elements change?

Friday, lecture 3: Bit patterns, full configuration interaction

Where we are

Thursday gave us the Hamiltonian in the form we shall use for the rest of the course,

$$\hat{H} = E_{\text{ref}} + \sum_{pq} f_{pq} \{a_p^\dagger a_q\} + \frac{1}{4} \sum_{pqrs} \langle pq | \hat{v} | rs \rangle \{a_p^\dagger a_q^\dagger a_s a_r\},$$

and the rules for its matrix elements between Slater determinants: non-zero only when the determinants differ by at most two orbitals.

Today:

- how a determinant and the action of $\hat{H}$ on it become **code**: bit patterns (section 3.16);
- **full configuration interaction**: expand the exact state in all determinants, diagonalise (chapter 5, sections 5.1–5.3);
- what the matrix looks like, and why it cannot be built for long: the block structure and the **exponential wall** (5.4–5.5);
- in the last 45 minutes: the pairing model by FCI, truncated CI, size consistency, the other methods as truncations, and the exercises (5.6–5.11).

From the formalism to a running code (section 3.16)

The time-consuming part of any shell-model or FCI code is the action of the Hamiltonian on a determinant,

$$\left(\sum_{pq} \langle p | \hat{h}_0 | q \rangle a_p^\dagger a_q + \frac{1}{4} \sum_{pqrs} \langle pq | \hat{v} | rs \rangle a_p^\dagger a_q^\dagger a_s a_r \right) a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger \cdots a_{\alpha_N}^\dagger |0\rangle.$$

A practically useful way to implement it: **encode a Slater determinant as a bit pattern**. With n single-particle states $\alpha_0, \dots, \alpha_{n-1}$ and $N \leq n$ particles, an occupied state is a 1, an empty one a 0. For $n = 16$ and particles in $\alpha_3, \alpha_6, \alpha_{10}, \alpha_{13}$:

$$\Phi_{3,6,10,13} = a_{\alpha_3}^\dagger a_{\alpha_6}^\dagger a_{\alpha_{10}}^\dagger a_{\alpha_{13}}^\dagger |0\rangle = |0001001000100100\rangle, \quad 2^3 + 2^6 + 2^{10} + 2^{13} = 9288.$$

- The ordering of the states is important: it defines the order of the creation operators, and hence the sign convention – the dictionary of section 3.1 with a fixed ordering.
- There are 2^n bit patterns but only $\binom{n}{N}$ determinants with N particles: $\dim \mathcal{H} = \binom{n}{N}$ – 1820 for $n = 16$, $N = 4$.
- The Fock-space matrices of the notebooks *are* this representation; today we never form the $2^n \times 2^n$ matrices, only act on one pattern at a time.

The action of a creation operator on a bit pattern

$a_{\alpha_4}^\dagger \Phi_{3,6,10,13} = a_{\alpha_4}^\dagger a_{\alpha_3}^\dagger a_{\alpha_6}^\dagger a_{\alpha_{10}}^\dagger a_{\alpha_{13}}^\dagger |0\rangle = -a_{\alpha_3}^\dagger a_{\alpha_4}^\dagger a_{\alpha_6}^\dagger a_{\alpha_{10}}^\dagger a_{\alpha_{13}}^\dagger |0\rangle = -|0001101000100100\rangle$,
one anticommutation to put $a_{\alpha_4}^\dagger$ in its place, while $a_{\alpha_6}^\dagger \Phi_{3,6,10,13} = -a_{\alpha_3}^\dagger (a_{\alpha_6}^\dagger)^2 a_{\alpha_{10}}^\dagger a_{\alpha_{13}}^\dagger |0\rangle = 0$.

The recipe

- If bit b_j is already 1 and we act with $a_{\alpha_j}^\dagger$: return the null vector (Pauli).
- If $b_j = 0$: set it to 1 and return the sign factor $(-1)^l$, l the number of bits set *before* bit j (the creation operators that $a_{\alpha_j}^\dagger$ has to pass).

For a_{α_j} : return zero if $b_j = 0$; otherwise clear the bit and return $(-1)^l$ with the same l .

$a_{\alpha_2}^\dagger 00111\rangle$	$=$	0	$a_{\alpha_2}^\dagger 10101\rangle$	$=$	0
$a_{\alpha_2}^\dagger 01011\rangle$	$=$	$(-1) 01111\rangle$	$a_{\alpha_2}^\dagger 10110\rangle$	$=$	0
$a_{\alpha_2}^\dagger 01101\rangle$	$=$	0	$a_{\alpha_2}^\dagger 11001\rangle$	$=$	$(+1) 11101\rangle$
$a_{\alpha_2}^\dagger 10011\rangle$	$=$	$(-1) 10111\rangle$	$a_{\alpha_2}^\dagger 11010\rangle$	$=$	$(+1) 11110\rangle$

(bits written $\alpha_0\alpha_1\alpha_2\alpha_3\alpha_4$ from left to right, as in the book).

The phase without a loop: bit masks

Consider $a_{\alpha_4}^\dagger a_{\alpha_0} \Phi_{0,3,6,10,13}$. Removing α_0 is a logical AND with the complement of the word $S_1 = |1000 \dots\rangle$; adding α_4 is a logical OR with $S_2 = |0001000 \dots\rangle$, after checking that the AND of S_2 with the state is zero. The phase of the pair move $a_{\alpha_j}^\dagger a_{\alpha_i}$ is

$$(-1)^{\#\{\text{occupied states strictly between } \alpha_i \text{ and } \alpha_j\}},$$

because the signs from removing α_i and from inserting α_j share all the occupied states outside the interval, which cancel in pairs. The mask of the states between α_i and α_j is integer arithmetic, the AND with the state selects the occupied ones, and a **bit-count** (popcount) gives l – here $|0001000 \dots\rangle$, $l = 1$, phase -1 .

```
def move(state, i, j): # a_j^dagger a_i acting on a bit pattern
    if not (state >> i) & 1: return 0, None # i empty: null vector
    if i == j: return 1, state # a_i^dagger a_i counts the occupation
    if (state >> j) & 1: return 0, None # j occupied: null vector
    lo, hi = min(i, j), max(i, j)
    between = state & ((1 << hi) - (1 << (lo + 1))) # occupied states strictly between
    phase = -1 if bin(between).count("1") % 2 else 1
    return phase, (state & ~(1 << i)) | (1 << j)
```

$\hat{H}|\Phi\rangle$ by bit operations: the inner loop of every FCI code

Apply the operator string of each term of $\hat{H}$ to the pattern, right to left, and collect {new pattern: amplitude}. Only *occupied* q, r, s contribute, so those loops run over the N occupied orbitals, not over all n :

```
def apply_hamiltonian(state, h, vAS):
    out = {}
    occ = [p for p in range(n) if (state >> p) & 1]
    for q in occ: # sum_pq h_pq a_p^+ a_q
        for p in range(n):
            sign, new = apply_string(state, [(p, True), (q, False)])
            if sign: out[new] = out.get(new, 0.0) + sign * h[p, q]
    for r in occ: # 1/4 sum <pq|v|rs> a_p^+ a_q^+ a_s a_r
        for s in occ:
            if s == r: continue
            for p in range(n):
                for q in range(n):
                    if vAS[p, q, r, s] != 0.0:
                        sign, new = apply_string(state, [(p, True), (q, True), (s, False), (r, False)])
                        if sign: out[new] = out.get(new, 0.0) + 0.25 * sign * vAS[p, q, r, s]
    return out
```

Cost per determinant $\mathcal{O}(Nn + N^2 n^2)$ bit operations, against $\binom{n}{N}$ inner products by matrices. Section 7 of the notebook compares with the Fock-space matrices for every pattern of a small space: identical, sign for sign; section 8 builds an FCI code on it.

Full configuration interaction: Slater determinants as a basis (section 5.1)

The simplest many-body wave function is a product, $\Psi \approx \phi_1(x_1) \cdots \phi_N(x_N)$, because we are really only good at thinking about one particle at a time. Products are easy – orthonormal, factorising matrix elements – and completely uncorrelated; correlations must be built up by **superposing many products**. This trade-off runs through the whole subject: a compact representation with difficult integrals, or easy integrals and very many states.

Antisymmetry turns the product into a determinant, chapter 3 the determinant into a string of creation operators. With the reference $|\Phi_0\rangle = \prod_{i \leq F} a_i^\dagger |0\rangle$ (our $|c\rangle$), the excited determinants are

$$|\Phi_i^a\rangle = a_a^\dagger a_i |\Phi_0\rangle, \quad |\Phi_{ij}^{ab}\rangle = a_a^\dagger a_b^\dagger a_j a_i |\Phi_0\rangle, \quad |\Phi_{ijk\dots}^{abc\dots}\rangle = a_a^\dagger a_b^\dagger a_c^\dagger \cdots a_k a_j a_i |\Phi_0\rangle,$$

the np - nh excitations or *configurations* of Thursday. Every determinant of N particles in n orbitals is one of these, for exactly one excitation level k : the number of its occupied orbitals not in $|\Phi_0\rangle$.

The idea of FCI

If the single-particle space describes the system well enough, diagonalising $\hat{H}$ in the space of *all* its determinants gives all eigenpairs exactly. FCI is the benchmark against which every approximation of the course (Hartree-Fock, TDA and RPA, perturbation theory, coupled cluster, neural-network and quantum-computing ansätze) is measured – which is why we treat it first.

The configuration-interaction expansion (section 5.2)

The exact ground state is an arbitrary superposition of all configurations,

$$|\Psi_0\rangle = C_0|\Phi_0\rangle + \sum_{ai} C_i^a |\Phi_i^a\rangle + \sum_{\substack{a<b \\ i<j}} C_{ij}^{ab} |\Phi_{ij}^{ab}\rangle + \dots = (C_0 + \hat{C})|\Phi_0\rangle,$$
$$\hat{C} = \sum_{ai} C_i^a a_a^\dagger a_i + \sum_{\substack{a<b \\ i<j}} C_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i + \dots$$

with the **correlation operator** $\hat{C}$ (restricted sums, every determinant once; unrestricted sums with antisymmetric C_{ij}^{ab} carry a factor $\frac{1}{4}$). With $C_0 \neq 0$ we may set $C_0 = 1$ (**intermediate normalisation**): $\langle\Psi_0|\Phi_0\rangle = 1$, $|\Psi_0\rangle = (1 + \hat{C})|\Phi_0\rangle$.

Compact notation: H a set of hole indices, P a set of particle indices, $\hat{A}_H^P$ the operator creating the excitation,

$$|\Psi_0\rangle = \sum_{PH} C_H^P |\Phi_H^P\rangle = \left(\sum_{PH} C_H^P \hat{A}_H^P \right) |\Phi_0\rangle, \quad E = \langle\Psi_0|\hat{H}|\Psi_0\rangle = \sum_{PP'HH'} C_H^{P*} \langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}$$

for unit normalisation $\sum_{PH} |C_H^P|^2 = 1$.

- If the expansion runs over *all* determinants of the chosen single-particle basis, this is **full configuration interaction**. The only approximation left is the truncation of the single-particle space itself.
- The C_i^a , C_{ij}^{ab} are what coupled-cluster theory will call amplitudes – with a multiplicative ansatz (lecture 4).

The eigenvalue problem (section 5.3)

Minimise the energy subject to normalisation, with a Lagrange multiplier λ : $\delta[\langle\Psi_0|\hat{H}|\Psi_0\rangle - \lambda\langle\Psi_0|\Psi_0\rangle] = 0$, i.e.

$$\sum_{PH} \sum_{P'H'} \left\{ \delta[C_H^{P*}] \langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'} + C_H^{P*} \langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle \delta[C_{H'}^{P'}] - \lambda(\delta[C_H^{P*}] C_{H'}^{P'} \delta_{PP'} \delta_{HH'} + \text{c.c.}) \right\} = 0.$$

Since $\delta[C^*]$ and $\delta[C]$ are complex conjugates it is necessary and sufficient that the coefficient of each $\delta[C_H^{P*}]$ vanishes; multiplying by C_H^{P*} and summing identifies $\lambda = E$:

The full CI equation

$$\sum_{P'H'} \langle\Phi_H^P|\hat{H} - E|\Phi_{H'}^{P'}\rangle C_{H'}^{P'} = 0 \quad \text{for every } PH$$

- The same result follows at once from $(\hat{H} - E)|\Psi_0\rangle = 0$ projected onto every $\langle\Phi_H^P|$.
- A Hermitian eigenvalue problem. Diagonalisation *is* the variational answer – Rayleigh-Ritz in the space of the chosen determinants: the lowest eigenvalue is the best upper bound to E_0 in that space, and it can only improve as the space grows.
- Matrix elements from the Condon-Slater rules of Thursday; diagonalisation by Householder-QL for small matrices, Lanczos for large ones – Lanczos needs only $\hat{H}|\Phi\rangle$, the bit code of three slides ago.

The structure of the Hamiltonian matrix (section 5.4)

Six fermions below the Fermi level, determinants ordered by excitation level:

	$0p0h$	$1p1h$	$2p2h$	$3p3h$	$4p4h$	$5p5h$	$6p6h$
$0p0h$	E_{ref}	f_{ai}	$\langle ab \hat{v} ij\rangle$	0	0	0	0
$1p1h$	$\times$	$\times$	$\times$	$\times$	0	0	0
$2p2h$	$\times$	$\times$	$\times$	$\times$	$\times$	0	0
$3p3h$	0	$\times$	$\times$	$\times$	$\times$	$\times$	0
$4p4h$	0	0	$\times$	$\times$	$\times$	$\times$	$\times$
$5p5h$	0	0	0	$\times$	$\times$	$\times$	$\times$
$6p6h$	0	0	0	0	$\times$	$\times$	$\times$

- The zeros are not accidental: a two-body operator connects determinants differing by at most two orbitals (Condon-Slater), and an $np-nh$ and an $mp-mh$ determinant differ by at least $|n - m|$ orbitals. The matrix is **banded in the excitation level, bandwidth two**, and overwhelmingly sparse.
- The first row is Thursday's Example 3: E_{ref} , f_{ai} , $\langle ab|\hat{v}|ij\rangle$, then nothing.
- A three-body force widens the band to three and multiplies the number of non-zero elements per row by $\sim N(n - N)$ – which is why three-body forces are normal-ordered with respect to $|\Phi_0\rangle$ and truncated at the two-body level (exercise 3b of this week's set).

In a Hartree-Fock basis: Brillouin's theorem empties one more block

If the single-particle states diagonalise the Fock matrix, $f_{ai} = 0$, and the reference does not couple to the singles at all:

	$0p0h$	$1p1h$	$2p2h$	$3p3h$	$4p4h$	$5p5h$	$6p6h$
$0p0h$	$\tilde{x}$	0	$\tilde{x}$	0	0	0	0
$1p1h$	0	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	0	0	0
$2p2h$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	0	0
$3p3h$	0	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	0
$4p4h$	0	0	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$
$5p5h$	0	0	0	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$
$6p6h$	0	0	0	0	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$

the tildes a reminder that the unitary transformation of the basis has changed every matrix element.

- The correlation energy is then carried entirely by the doubles at leading order – the starting point of both perturbation theory and coupled-cluster theory.
- For the pairing model every even-odd block is empty, whatever the basis: the interaction moves pairs and cannot break one, so the number of broken pairs (the seniority) is conserved. The symmetry does more than Brillouin's theorem would (lecture 4).

Pause to think: counting determinants

Pause to think #5

(about 4 min – discuss with your neighbour)

N particles in n single-particle states, reference $|\Phi_0\rangle$.

1. How many determinants are k particle-hole excitations of $|\Phi_0\rangle$? Check that they add up to $\binom{n}{N}$.
2. Evaluate for $n = 8$, $N = 4$ (the pairing model of Thursday's Example 5). What is the dimension of the singles block, and of the space kept in a “singles and doubles” truncation?

Answer to pause #5

1. Choose which k of the N occupied orbitals to empty, $\binom{N}{k}$ ways, and which k of the $n - N$ empty ones to fill, $\binom{n-N}{k}$ ways: $\binom{N}{k} \binom{n-N}{k}$ determinants at level k . Every N -particle determinant arises exactly once, so
$$\sum_k \binom{N}{k} \binom{n-N}{k} = \sum_k \binom{N}{N-k} \binom{n-N}{k} = \binom{n}{N} - \text{Vandermonde's identity.}$$
2. $\binom{4}{k}^2 = 1, 16, 36, 16, 1$ for $k = 0, \dots, 4$, total $70 = \binom{8}{4}$. The singles block is 16×16 ; singles and doubles keep $1 + 16 + 36 = 53$ of the 70 determinants – and, as we shall see, essentially all of the correlation energy of the pairing model at small g .

The exponential wall (section 5.5)

Full configuration interaction is exact, and that is precisely its problem. $\dim \mathcal{H} = \binom{n}{N}$ is the number of bit patterns with N bits set out of n , and at half filling Stirling's formula gives

$$\binom{n}{n/2} \sim \frac{2^n}{\sqrt{\pi n/2}} : \quad \text{the dimension \textcolor{brown}{doubles} for every single-particle state added.}$$

n	$\binom{n}{n/2}$	$2^n / \sqrt{\pi n/2}$	landmark
10	2.520×10^2	2.584×10^2	a toy model
20	1.848×10^5	1.871×10^5	limit of dense diagonalisation ($\sim 10^5$)
36–38	9.08×10^9 – 3.53×10^{10}		limit of Lanczos on a large machine ($\sim 10^{10}$)
40	1.378×10^{11}	1.387×10^{11}	
80	1.075×10^{23}	1.078×10^{23}	Avogadro's number of amplitudes
160	9.205×10^{46}	9.219×10^{46}	

- At fixed N , $\binom{n}{N} \sim n^N / N!$ is a polynomial: adding orbitals to a few valence particles is cheap – calculations near a closed shell are much easier than in the middle of an open one. The half-filled line is the envelope of the family (section 9 of the notebook).
- **A realistic example.** The eight neutrons of ^{16}O in the four lowest major oscillator shells ($2 + 6 + 12 + 20 = 40$ states): $\binom{40}{8} = 7.7 \times 10^7$, inside the Lanczos band. Protons independently: $\binom{40}{8}^2 = 5.9 \times 10^{15}$. Symmetries cut this by orders of magnitude; one more shell restores the growth at once.

The same wall in the models of the course, and why it is not the end of the story

Lipkin: $\binom{2N}{N}$ determinants, of which $SU(2)$ isolates $N + 1$ – a symmetry accident. *Pairing*: $\binom{2n}{N}$ determinants, $\binom{n}{N/2}$ of seniority zero. *Hubbard*: $\binom{L}{N_{\uparrow}}\binom{L}{N_{\downarrow}}$; a spin chain has 2^L by construction. A conservation law removes a factor, never the exponential.

Many-body connection

The dimension of $\mathcal{H}$ counts the coefficients an *arbitrary* state would need. Physical ground states are not arbitrary: for local, gapped Hamiltonians the entanglement across a cut grows with its area, not its volume, and such states need a polynomial number of parameters – matrix product states, or a neural-network quantum state that replaces the $\binom{n}{N}$ amplitudes by a function of the occupation pattern. Full CI is the reference against which all such compressions are measured.

Quantum computing connection

A register of n qubits has dimension 2^n by construction – the same growth, as a resource rather than a cost: the bit pattern of a determinant is a basis state of the register, and the FCI ground state a state of the register. How fermion operators become qubit operators is the subject of FYS5419/9419 in the spring.

Summary of lecture 3, and what comes next

- A Slater determinant is an integer; $a^\dagger$ and a are bit operations with a popcount for the sign; $\hat{H}|\Phi\rangle$ is a loop over the occupied orbitals and the non-zero matrix elements.
- FCI: $|\Psi_0\rangle = (1 + \hat{C})|\Phi_0\rangle$ over all configurations; the variational principle turns the expansion into the eigenvalue problem $\sum_{P'H'} \langle \Phi_H^P | \hat{H} - E | \Phi_{H'}^{P'} \rangle C_{H'}^{P'} = 0$.
- The matrix is banded in the excitation level (bandwidth two), its first row is $E_{\text{ref}}, f_{ai}, \langle ab|\hat{v}|ij\rangle$; a Hartree-Fock basis empties the $0p0h-1p1h$ block.
- $\dim \mathcal{H} = \binom{n}{N}$, exponential at fixed filling: dense diagonalisation to $\sim 10^5$, Lanczos to $\sim 10^{10}$, beyond that the expansion itself must be truncated.

Many-body connection

In the last 45 minutes we run the machinery on the pairing model – 70 determinants, 6 of seniority zero, one ground-state energy – and then *truncate* the expansion, which is where every other method of the course begins: CIS, CID, CISD and their failure of size consistency, and the exponential ansatz that repairs it.

Friday, lecture 4: FCI for the pairing model, exercises

Full CI for the pairing model (section 5.6)

The pairing Hamiltonian of Thursday's Example 5, four doubly degenerate levels, four particles, $\xi = 1$:

$$\hat{H} = \xi \sum_{p\sigma} (p-1) a_{p\sigma}^\dagger a_{p\sigma} - \frac{g}{2} \sum_{pq} a_{p+}^\dagger a_{p-}^\dagger a_{q-} a_{q+}.$$

The full space has $\binom{8}{4} = 70$ determinants, 1, 16, 36, 16, 1 over the excitation levels, and the block table computed by `fci.py` (section 8 of the notebook) is

	0p0h	1p1h	2p2h	3p3h	4p4h
0p0h	×	0	×	0	0
1p1h	0	×	0	×	0
2p2h	×	0	×	0	×
3p3h	0	×	0	×	0
4p4h	0	0	×	0	×

- The even-odd blocks are empty as well: the interaction moves particles two at a time and never changes the number of broken pairs. The **seniority** s (number of unpaired particles) commutes with $\hat{H}$: a six-dimensional $s = 0$ block, an $s = 2$ block, an $s = 4$ block.
- The ideal first test: small enough for the full space, and the seniority-zero answer of chapter 4 is known.

The seniority-zero matrix, by the rules of Thursday

Label a seniority-zero configuration by the pair of occupied levels: $|12\rangle, |13\rangle, |14\rangle, |23\rangle, |24\rangle, |34\rangle$. The diagonal element of $|pq\rangle$ is $2\xi[(p-1) + (q-1)] - g$ (each level holds two particles; the $p = q$ terms of the interaction give $-g/2$ for each of the two pairs – Example 5 with the reference $|12\rangle$ and the $2p2h$ state $|23\rangle$). One application of the interaction moves *one* pair, so two configurations are connected by $-g/2$ if and only if they share exactly one level:

$$H = \begin{pmatrix} 2 - g & -g/2 & -g/2 & -g/2 & -g/2 & 0 \\ -g/2 & 4 - g & -g/2 & -g/2 & 0 & -g/2 \\ -g/2 & -g/2 & 6 - g & 0 & -g/2 & -g/2 \\ -g/2 & -g/2 & 0 & 6 - g & -g/2 & -g/2 \\ -g/2 & 0 & -g/2 & -g/2 & 8 - g & -g/2 \\ 0 & -g/2 & -g/2 & -g/2 & -g/2 & 10 - g \end{pmatrix}$$

- Every entry is a Condon-Slater element: $\langle \Phi_0 | \hat{H} | \Phi_0 \rangle = E_{\text{ref}} = 2 - g$, $\langle \Phi_{ij}^{ab} | \hat{H} | \Phi_0 \rangle = \langle ab | \hat{v} | ij \rangle = -g/2$ for a pair moved as a whole, and $|12\rangle - |34\rangle$ differ by *four* orbitals: zero.
- The three unconnected pairs, $|12\rangle - |34\rangle$, $|13\rangle - |24\rangle$, $|14\rangle - |23\rangle$, are the zeros in the matrix.

Results: 70×70 against 6×6

g	E_{ref}	E_0 (6×6)	E_0 (70×70)	broken-pair amplitude
-1.00	3.000000	2.77987014	2.77987014	$< 10^{-15}$
-0.50	2.500000	2.43688426	2.43688426	$< 10^{-15}$
0.00	2.000000	2.00000000	2.00000000	0
0.50	1.500000	1.41677428	1.41677428	$< 10^{-15}$
1.00	1.000000	0.63554847	0.63554847	$< 10^{-15}$

- Identical to every digit. The 64 determinants with a broken pair are present in the basis but carry no amplitude in the ground state: the symmetry that chapter 4 imposed by hand is **recovered automatically** by the diagonalisation.
- This is the general situation. A symmetry of $\hat{H}$ block-diagonalises the FCI matrix, and one may exploit it (a smaller matrix, a labelled spectrum) or ignore it (a simpler code, a larger matrix). Production codes exploit it; `fci.py` does not, so that the two answers can be compared.
- The correlation energy $E_0 - E_{\text{ref}}$ is -0.083 at $g = 0.5$ and -0.364 at $g = 1$: 36% of the reference energy. At small $|g|$ it is quadratic, and the coefficient can be computed by hand (pause #6).

Pause to think: the correlation energy at small g

Pause to think #6

(about 4 min – discuss with your neighbour)

With the reference $|12\rangle$, second-order perturbation theory gives

$$E^{(2)} = \sum_{\Phi \neq \Phi_0} \frac{|\langle \Phi_0 | \hat{H}_I | \Phi \rangle|^2}{E_0^{(0)} - E_\Phi^{(0)}}.$$

1. Which configurations contribute, with what matrix element and which unperturbed excitation energies? Evaluate $E^{(2)}$.
2. Compare with the exact correlation energies -0.01954 ($g = 0.25$), -0.0832 ($g = 0.5$) and -0.3645 ($g = 1$). What do you conclude about the higher orders?

Answer to pause #6

1. The four configurations reachable by moving one pair, $|13\rangle, |14\rangle, |23\rangle, |24\rangle$, each with matrix element $-g/2$ and unperturbed excitation energies 2, 4, 4, 6:

$$E^{(2)} = \frac{g^2}{4} \left(\frac{1}{-2} + \frac{1}{-4} + \frac{1}{-4} + \frac{1}{-6} \right) = -\frac{7}{24} g^2 = -0.2917 g^2.$$

2. -0.01823 against -0.01954 at $g = 0.25$, -0.0729 against -0.0832 at $g = 0.5$, -0.2917 against -0.3645 at $g = 1$: the leading order is right at small coupling and misses 20% at $g = 1$. The higher orders are negative for $g > 0$ and their sum is what FCI delivers; perturbation theory (later in the course) computes them one order at a time, FCI all at once.

Truncating the expansion (section 5.8)

Since the full expansion is out of reach for all but the smallest systems, we truncate it by excitation level:

$$|\Psi_{\text{CIS}}\rangle = (1 + \hat{C}_1)|\Phi_0\rangle, \quad |\Psi_{\text{CID}}\rangle = (1 + \hat{C}_2)|\Phi_0\rangle, \quad |\Psi_{\text{CISD}}\rangle = (1 + \hat{C}_1 + \hat{C}_2)|\Phi_0\rangle, \dots$$

up to CISDTQ and, at $\hat{C}_N$, full CI. Each is variational in a subspace: upper bounds to E_0 , decreasing monotonically as the truncation is relaxed. The space at level k holds $\binom{N}{k} \binom{n-N}{k}$ determinants, polynomial of degree $2k$, and the work grows roughly as n^{2k+2} : CISD at $\mathcal{O}(n^6)$ is affordable, CISDTQ at $\mathcal{O}(n^{10})$ is not.

g	E_{ref}	CIS (17)	CID (37)	CISD (53)	FCI (70)
0.25	1.750000	1.75000000	1.73050700	1.73050700	1.73045985
0.50	1.500000	1.50000000	1.41759645	1.41759645	1.41677428
1.00	1.000000	1.00000000	0.64890655	0.64890655	0.63554847
2.00	0.000000	0.00000000	-1.35208670	-1.35208670	-1.48965216

- The singles change nothing (a single excitation breaks a pair – the reference is decoupled from the whole singles block), and CISD coincides with CID.
- The doubles carry 99.8% of the correlation energy at $g = 0.25$, 96.3% at $g = 1$, only 90.8% at $g = 2$: truncation at fixed level works while the reference is good and degrades as the coupling grows. In the Hubbard ring (section 5.7) the same pattern: 98.7% at $U/t = 1$, 72% at $U/t = 8$ – the signature of **strong correlation**.

Size consistency (section 5.9)

There is a defect of truncated CI more serious than the loss of accuracy, because it does not go away as the basis is improved. Two identical systems A and B , infinitely far apart: $\hat{H} = \hat{H}_A + \hat{H}_B$, the exact ground state is the product $|\Psi_A\rangle|\Psi_B\rangle$, and

$$E(AB) = E(A) + E(B) = 2E(A) \quad (\text{size consistency}).$$

Full CI passes, since the product state lies in the full space. Truncated CI does not: if A and B are each doubly excited, the combined state is a *quadruple* excitation of the combined reference, and CID has no room for it. The product of two doubles is a quadruple – truncation at a fixed level is not multiplicatively closed.

g	$2E(A)$	$E(AB)$, FCI	$E(AB)$, CID	CID error
0.25	-0.26556444	-0.26556444	-0.26550480	5.96×10^{-5}
0.50	-0.56155281	-0.56155281	-0.56066017	8.93×10^{-4}
1.00	-1.23606798	-1.23606798	-1.22474487	1.13×10^{-2}

Two non-interacting copies of a two-level, one-pair system (section 8 of the notebook). With M subsystems the error grows with M : the energy per particle of truncated CI drifts with the size of the system – fatal for nuclear matter, the electron gas, a large molecule.

The repair: make the ansatz multiplicative

For two non-interacting one-pair systems the exact state is a product,

$$(1 + c\hat{A}_2^A)(1 + c\hat{A}_2^B)|\Phi_0\rangle = (1 + c\hat{A}_2^A + c\hat{A}_2^B + c^2\hat{A}_2^A\hat{A}_2^B)|\Phi_0\rangle,$$

and the last term – both subsystems excited at once – is a $4p4h$ excitation of the combined reference, absent from CID, with a coefficient that is *not* independent: c^2 , the product of the two doubles.

Coupled-cluster theory in one line

$$|\Psi_{\text{CCSD}}\rangle = e^{\hat{T}_1 + \hat{T}_2}|\Phi_0\rangle = (1 + \hat{T}_1 + \hat{T}_2 + \frac{1}{2}\hat{T}_2^2 + \dots)|\Phi_0\rangle$$

The quadruple $\frac{1}{2}\hat{T}_2^2$ appears automatically with its coefficient fixed by the doubles; for non-interacting subsystems $e^{\hat{T}} = e^{\hat{T}_A}e^{\hat{T}_B}$ and size consistency is exact – with the *same* number of parameters as CISD.

- For the two copies, $\hat{T}_2 = t(\hat{A}_2^A + \hat{A}_2^B)$ with commuting pieces and $(\hat{A}_2^X)^2 = 0$ gives $e^{\hat{T}_2} = (1 + t\hat{A}_2^A)(1 + t\hat{A}_2^B)$ exactly (chapter 5, exercise 4 of the lecture notes).
- This is why coupled cluster, not truncated CI, is the workhorse of modern many-body physics; we develop it properly later in the course.

Truncations and the other many-body methods (section 5.10)

Project $(\hat{H} - E)|\Psi_0\rangle = 0$ onto $\langle\Phi_0|$ and onto $\langle\Phi_H^P|$:

$$E = \langle\Phi_0|\hat{H}|\Phi_0\rangle + \sum_{P'H'\neq 0} \langle\Phi_0|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}, \quad (E - \langle\Phi_H^P|\hat{H}|\Phi_H^P\rangle) C_H^P = \sum_{P'H'\neq PH} \langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}.$$

The first: the correlation energy comes from the coupling of the reference to singles and doubles only, $\sum_{ai} f_{ai} C_i^a + \sum_{a<b, i<j} \langle ab|\hat{v}|ij\rangle C_{ij}^{ab}$ (Example 3). The second: coupled equations for the coefficients, solvable by iteration. Every method of the course is a prescription for which terms to keep and how to solve it:

- **Hartree-Fock**: keep only C_0 and vary the *basis* so that $f_{ai} = 0$ – optimise the reference, not the expansion (end of next week).
- **Tamm-Dancoff**: diagonalise the $1p1h$ block alone (CIS for excited states; the matrix of Example 4). **RPA** adds $2p2h$ correlations to the ground state.
- **Perturbation theory**: expand in the interaction – $E \rightarrow E_{\text{ref}}$, $C \rightarrow 0$ on the right gives the first-order coefficients, substituting back the second-order energy (pause #6), and so on.
- **Coupled cluster**: the truncation of CISD with the multiplicative ansatz; the nonlinear terms restore size consistency.

One question, several answers

Which parts of the CI expansion do we keep, and additively or multiplicatively? Full CI is the fixed point that measures each answer: whenever a model is small enough to solve exactly, solve it and calibrate.

Practical considerations (section 5.11)

- **Never store the matrix.** The FCI matrix is sparse and banded, and for interesting dimensions cannot be stored at all. Lanczos needs only $\hat{H}v$, computed by looping over the determinants and the non-zero elements with the bit representation – `apply_hamiltonian` of lecture 3. The single most important structural choice in an FCI code. (Chapter 5, exercise 6: 10^6 determinants with 10^3 non-zero elements per row take 12 GB sparse, 8 TB dense.)
- **Reorthogonalise.** Lanczos vectors lose orthogonality as soon as an eigenvalue has converged and spurious copies appear; full or selective reorthogonalisation, at the price of storing the vectors.
- **Use the symmetries.** Angular momentum, parity, particle number, total momentum and spin projection commute with $\hat{H}$ and block-diagonalise the matrix: orders of magnitude in dimension, and labelled states – 70 determinants without the symmetry today, 6 with it, the same energy.

Many-body connection

Large-scale shell-model codes (nuclear physics) and FCI codes (quantum chemistry) are exactly this: bit patterns, a symmetry-adapted basis, and Lanczos on a matrix never written down. Dimensions of 10^{10} are reached on large machines; beyond that the wall stands, and the rest of the course is about climbing it approximately.

Exercise session: week 37

The exercises are taken from the chapter exercise sections of the lecture notes, where full answers are given: chapter 3, exercise 4, and chapter 5, exercises 1–3.

Exercise 1 (chapter 3, exercise 4): normal ordering with respect to a Fermi vacuum

- a) With $|c\rangle$ the reference determinant, evaluate the four contractions of a_p and $a_p^\dagger$ with respect to $|c\rangle$ and say which survive.
- b) Use them to compute $\langle c|\hat{H}_0|c\rangle$ for $\hat{H}_0 = \sum_{pq} \langle p|\hat{h}_0|q\rangle a_p^\dagger a_q$.
- c) Explain why the same argument fails if $|c\rangle$ is replaced by a BCS state.
- d) (Added.) Repeat the derivation of Thursday for $\hat{V}$ and verify the six terms of the four-operator expansion, with their signs, by the crossing rule.

Exercise session: week 37 (continued)

Exercise 2 (chapter 5, exercise 1): counting determinants

- a) Show that the number of k particle-hole excitations of a given reference is $\binom{N}{k} \binom{n-N}{k}$, and verify $\sum_k \binom{N}{k} \binom{n-N}{k} = \binom{n}{N}$.
- b) Evaluate this for $n = 8$, $N = 4$ and compare with the distribution 1, 16, 36, 16, 1.
- c) Use Stirling's formula to derive $\binom{n}{n/2} \sim 2^n / \sqrt{\pi n/2}$, and determine the largest n for which $\binom{n}{n/2}$ stays below 10^{10} .

Exercise 3 (chapter 5, exercise 2): which blocks vanish

Let $\hat{H} = \hat{H}_0 + \hat{H}_I$ with $\hat{H}_I$ a two-body operator.

- a) Show that $\langle \Phi_H^P | \hat{H} | \Phi_{H'}^{P'} \rangle = 0$ whenever the two determinants differ by more than two single-particle states, and deduce the band structure of the FCI matrix.
- b) Suppose $\hat{H}$ also contains a genuine three-body force. How does the band structure change, and what does that do to the cost of a matrix-vector product?
- c) For the pairing Hamiltonian, show that every block connecting an even to an odd excitation level vanishes.

Exercise session: week 37 (continued)

Exercise 4 (chapter 5, exercise 3): full CI for the pairing model

Four doubly degenerate levels with $\varepsilon_p = (p - 1)\xi$, four particles, $\xi = 1$.

- Set up the six seniority-zero configurations and write down the 6×6 Hamiltonian matrix – with the Condon-Slater rules, not by inspection.
- Diagonalise it for $g = -1, -0.5, 0, 0.5, 1$ and reproduce the table of lecture 4.
- Repeat the calculation in the full 70-dimensional space (`fci.py`, or section 8 of the notebook) and confirm that the ground-state energy is unchanged. Inspect the eigenvector and verify that every determinant with a broken pair has zero amplitude.
- Plot the correlation energy against g and show that it is quadratic for small $|g|$, with the coefficient $-7/24$ of pause #6.

Hints: for 3a) count the operators of the bra, the ket and $\hat{H}_I$ and ask whether the generalised theorem can contract them all; for 4a) every entry is one of E_{ref} , $\langle \Phi_{ij}^{ab} | \hat{H} | \Phi_0 \rangle = \langle ab | \hat{v} | ij \rangle$, or the $2p2h$ diagonal formula of Thursday's Example 4.

What to do with the notebook

WeeklySlides/week37.ipynb contains, in this order:

1. Fermion operators as matrices on a six-orbital Fock space, and a random Hamiltonian (reminder).
2. The generalised theorem as an operator identity; the four-term one-body formula, $\langle k\ell|\hat{H}_I|ij\rangle = \langle k\ell|v|ij\rangle_{AS}$, and the sparsity rule.
3. Section 3.4: $|c\rangle$, the b operators, $\hat{N}$ and $\hat{N}_{qp}$, the two Fermi-vacuum contractions – every pair of operators checked.
4. Section 3.5: normal ordering relative to $|c\rangle$ as code; Wick's theorem relative to $|c\rangle$ on random strings; the six-term expansion of $a_p^\dagger a_q^\dagger a_s a_r$ for all 6^4 index choices; $\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$ as a matrix identity.
5. Examples 3 and 4: f_{ai} , $\langle ab|\hat{V}|ij\rangle$, zero for triples, the $1p1h$ and $2p2h$ energies, the TDA matrix, and the particle/hole blocks of $\hat{F}_N$ and $\hat{V}_N$ (the five groups, with the derived prefactors).
6. Example 5: the pairing model, every number of the table.
7. Section 3.16: bit patterns, and $\hat{H}|\Phi\rangle$ by bit operations against the matrices.
8. Chapter 5: a small FCI code (SlaterBasis); the 70-dimensional pairing calculation, the block table, the comparison with the 6×6 matrix, the broken-pair amplitudes, CIS/CID/CISD against FCI, and the size-consistency test.
9. The exponential wall: the table and the plot of $\binom{n}{N}$ against n .

Play with it: change the Fermi level, the number of orbitals, the coupling g ; try the reference with levels 1 and 3 filled; run the FCI code for six levels and six particles (924 determinants) and reproduce Figure 5.2 of the lecture notes.

Summary of week 37

The formalism

- Generalised Wick: only inter-group lines; Condon-Slater rules in four surviving pairings.
- Fermi sea as vacuum: $b_a^\dagger = a_a^\dagger$, $b_i^\dagger = a_i$; contractions $a_a a_b^\dagger \rightarrow \delta_{ab}$, $a_i^\dagger a_j \rightarrow \delta_{ij}$.
- $\hat{H} = E_{\text{ref}} + \sum f_{pq} \{a_p^\dagger a_q\} + \frac{1}{4} \sum \langle pq|v|rs \rangle \{a_p^\dagger a_q^\dagger a_s a_r\}$, with $f_{pq} = h_{pq} + \sum_i \langle pi|v|qi \rangle$ and $E_{\text{ref}} = \sum_i h_{ii} + \frac{1}{2} \sum_{ij} \langle ij|v|ij \rangle$.
- First row of $\hat{H}$: E_{ref} , f_{ai} , $\langle ab|v|ij \rangle$, 0.

In code: a determinant is an integer, $\hat{H}|\Phi\rangle$ a loop over occupied orbitals, and FCI is `numpy.linalg.eigh` of a matrix one never stores for real.

Full configuration interaction

- $|\Psi_0\rangle = (1 + \hat{C})|\Phi_0\rangle$; the variational principle gives the eigenvalue problem.
- Banded matrix, bandwidth two; symmetries block-diagonalise it further.
- $\dim \mathcal{H} = \binom{n}{N} \sim 2^n / \sqrt{n}$: the exponential wall.
- Pairing model: 70 = 6 to every digit; CID 96% at $g = 1$; truncated CI is not size consistent, $e^{\hat{T}}$ is.

Next week, and a closing remark

Next week (week 38)

Full configuration interaction continued, with worked examples: the FCI code on the pairing model beyond four levels (six levels, six particles, 924 determinants, CISD and CISDTQ against FCI), the Hubbard ring in the momentum basis (section 5.7) and its strong-correlation regime, the amplitude equations solved by iteration, and the counting exercises done in class. At the end of the week we begin Hartree-Fock theory (chapter 6): the reference determinant as a variational ansatz, the energy functional E_{ref} of this week and its minimisation. Reading: chapter 5 (all sections) and the first sections of chapter 6; Szabo and Ostlund chapters 3–4.

Three weeks ago every expectation value cost a page of permutations. This week $\langle c | \hat{H} | c \rangle$, f_{ai} and $\langle ab | \hat{v} | ij \rangle$ were read off from a normal-ordered Hamiltonian, the whole Hamiltonian matrix of a model was written down from three rules, and diagonalising it gave the exact answer – for 70 determinants. The exponential wall says that this cannot last, and the truncations that it forces upon us are precisely the approximations that the following weeks elevate into methods: Hartree-Fock empties one block, perturbation theory iterates the amplitude equations, and coupled cluster makes the ansatz multiplicative.